

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

**(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)**

Department of Computer Engineering



Project Report on

SecureX AI - Enabling Decisions, Protecting Data

In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology
(B.Tech.) Degree in Computer Engineering at the University of Mumbai
Academic Year 2025-26

Project Mentor

Mrs. Yugchhaya Galphat

Submitted by

- 1) Niraj Chaudhari, D12B -11
- 2) Shubham Mishra, D12B -35
- 3) Tanmay Jadhav, D12B -20
- 4) Gauri Doiphode, D12B -16

(2025-26)

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering



CERTIFICATE of Approval

This is to certify that *Niraj Chaudhari (D12B -11), Shubham Mishra (D12B -35), Tanmay Jadhav (D12B -20), Gauri Doiphode (D12B -16)* of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on “*SecureX AI — Enabling Decisions, Protecting Data*” as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of *Mrs. Yugchhaya Galphat* in the year 2025-26.

Date

Internal Examiner

External Examiner

Project Mentor

Head of the Department

Dr. Mrs. Nupur Giri

Principal

Dr. J. M. Nair

ACKNOWLEDGEMENT

We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

It gives us immense pleasure to express our deep and sincere gratitude to Professor **Mrs. Yugchhaya Galphat** for her kind help and valuable advice during the development of project synopsis and for her guidance and suggestions.

We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair** , for giving us this valuable opportunity to do this project.

We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

We convey our deep sense of gratitude to all teaching and non-teaching staff for their constant encouragement, support and selfless help throughout the project work. It is a great pleasure to acknowledge the help and suggestion, which we received from the Department of Computer Engineering.

We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR CAPSTONE PROJECT

Learners will be to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

The proposed system presents a secure, organization-centric artificial intelligence platform designed to facilitate informed decision-making while maintaining strict standards of data privacy, security, and accountability. With the growing adoption of generative AI across industries, existing solutions largely remain consumer-oriented and lack essential enterprise features such as controlled access, regulatory compliance, and data governance. This creates significant challenges for organizations handling sensitive information.

To address these limitations, the system incorporates role-based access control (RBAC) to ensure that information access and generated outputs are aligned with user permissions. It further introduces domain-specific templates tailored to functional areas such as Human Resources, Finance, and Legal, enabling structured and context-aware interactions. In addition, a real-time compliance mechanism is integrated to prevent unauthorized exposure of confidential data.

The platform is built upon a Retrieval-Augmented Generation (RAG) framework combined with policy-driven filtering, ensuring that responses are grounded exclusively in authorized internal data sources. By operating within a controlled organizational environment, the system enhances reliability, transparency, and auditability of AI-driven outputs.

Overall, the proposed solution provides a practical and scalable approach to enterprise AI adoption, reducing operational overheads, minimizing manual effort, and eliminating the need for extensive in-house AI infrastructure, while maintaining compliance with organizational and regulatory requirements.

INDEX

Chapter No.	Title	Page No.
1	Introduction	1-7
1.1	Introduction to the project	1
1.2	Motivation for the project	1
1.3	Drawback of the existing system	2
1.4	Problem Definition	4
1.5	Relevance of the Project	5
1.6	Methodology used	6
2	Literature Survey	8-14
2.1	Research Papers – Abstract of the research paper; Inference drawn from the paper	8
2.2	Books / Articles / Newspaper referred	10
2.3	Patent Search – Study of 3–4 patents; Title and year of patent; Summary of patent; European Patent Link: http://worldwide.espacenet.com/ ; US Patent Link	14
3	Requirement of Proposed System	15-17
3.1	Functional Requirements	15
3.2	Non-Functional Requirements	16
3.3	Constraints	16
3.4	Hardware & Software Requirements	16
3.5	Techniques utilized till date for the proposed system	17
3.6	Tools utilized till date for the proposed system	17
3.7	SDGs mapped with the proposed project and justification	17

4	Proposed Design	18-31
4.1	Block diagram representation of the proposed system with explanation	19
4.2	Modular diagram representation of the proposed system with explanation	21
4.3	Design of the proposed system – Data Flow / Use Case Diagram; Flowchart; State Transition / Activity Diagram; ER Diagram; Screenshot of implementation	23
4.4	Algorithms utilized in the existing systems	27
4.5	Project scheduling and tracking using Timeline / Gantt Chart	30
5	Implementation Results and Discussions	32-35
5.1	Data Collection / Acquisition	32
5.2	Data Preprocessing Techniques	33
5.3	Experimental Setup	33
5.4	Evaluation Metrics	34
5.5	Gap analysis of existing systems	35
6	Plan of Action for the Next Semester	36-38
6.1	Work done till date	36
6.2	Plan of action for Project II	37
7	Conclusion	39
8	References (IEEE format)	40
9	Appendix	41
9.1	List of Figures	41
9.2	Xerox of Project Review 1 & 2 Evaluation sheet	42

1. INTRODUCTION

1.1 Introduction to the project

The project titled *SecureX AI — Enabling Decisions, Protecting Data* focuses on developing a secure, enterprise-oriented generative AI platform that supports intelligent decision-making while maintaining strict control over organizational data. With the rapid growth of Large Language Models (LLMs), organizations are increasingly adopting AI technologies to enhance productivity, automate workflows, and extract insights from large volumes of data.

However, existing AI tools are primarily designed for general-purpose use and do not provide mechanisms to enforce organizational policies such as role-based access, data governance, and compliance requirements. This project addresses these limitations by introducing a controlled AI system that operates within an organization's environment and interacts only with approved internal data sources.

The system is built using a Retrieval-Augmented Generation (RAG) architecture, which ensures that responses are generated based on verified internal documents rather than generic knowledge. It also integrates Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) to restrict access based on user roles and permissions. Additionally, a Semantic Verification Layer is implemented to ensure the accuracy and reliability of responses by validating them against source data.

The platform is developed using technologies such as Python, FastAPI, React, and vector databases like FAISS or Pinecone, making it scalable and suitable for deployment in enterprise environments.

1.2 Motivation for the project

The motivation behind this project arises from the increasing need for secure and reliable AI systems in enterprise environments. Organizations today deal with large volumes of sensitive and structured data that can be effectively utilized through AI for better decision-making and operational efficiency.

Despite the advantages of AI, current solutions lack essential enterprise features such as access control, auditability, and compliance enforcement. This creates hesitation in adopting AI systems for critical operations. Additionally, building custom AI infrastructure is expensive and resource-intensive, especially for small and medium-sized enterprises.

Therefore, there is a strong need for a system that enables organizations to leverage AI capabilities while ensuring security, controlled access, and compliance. This project aims to provide a cost-effective and scalable solution that bridges this gap and promotes safe AI adoption across industries.

1.3 Drawback of the existing system

The existing AI systems, especially general-purpose and consumer-oriented tools, face several critical limitations when applied to enterprise environments. These limitations restrict their usability, reliability, and adoption in organizations dealing with sensitive and structured data.

1. Lack of Access Control

One of the major drawbacks of current AI systems is the absence of proper access control mechanisms. Most tools do not support Role-Based Access Control (RBAC) or Attribute-Based Access Control (ABAC), which are essential in enterprise environments.

In an organization, different users such as employees, managers, and administrators have different levels of access rights. However, existing systems treat all users equally, which can result in unauthorized access to confidential information. This lack of controlled access increases the risk of internal data misuse and makes such systems unsuitable for enterprise deployment.

2. Compliance Limitations

Organizations are required to follow strict regulatory standards such as GDPR, HIPAA, and Digital Personal Data Protection Act. Existing AI systems do not provide built-in mechanisms to ensure compliance with these regulations.

There is no guarantee that sensitive data is handled, processed, or stored according to legal requirements. This creates serious legal and financial risks for organizations, including penalties and loss of trust. As a result, enterprises hesitate to adopt such systems for handling critical data.

3. No Auditability and Traceability

Another significant limitation is the lack of auditability. Current AI systems do not maintain detailed logs of user actions, data access, or response generation processes.

In enterprise environments, it is important to track:

- Who accessed what data
- When the data was accessed
- How a particular response was generated

Without proper logging and traceability, it becomes difficult to investigate errors, ensure accountability, or verify the correctness of outputs. This reduces trust in the system and limits its use in decision-making processes.

4. Hallucination Issues

AI models often suffer from the problem of hallucination, where they generate responses that are incorrect, misleading, or not based on actual data. This occurs because the models rely on general training data and lack domain-specific context.

In enterprise scenarios, even small inaccuracies can lead to wrong decisions, financial losses, or compliance violations. Since existing systems do not validate responses against trusted internal data sources, the reliability of their outputs becomes questionable.

5. High Implementation Cost

Building a secure and customized AI solution from scratch requires significant investment in infrastructure, skilled professionals, and ongoing maintenance. Organizations need to manage:

- Data storage and processing systems
- Model training and fine-tuning
- Security and compliance mechanisms

For many small and medium-sized enterprises, these costs are too high, making it difficult for them to adopt advanced AI solutions. As a result, they either avoid using AI or rely on inadequate tools that do not meet their requirements.

1.4 Problem Definition

In our project, we are dealing with the issue that organizations generate a huge amount of internal data such as documents, reports, policies, and records, but this data is not used effectively for decision-making. The main reason is that there is no proper system that allows safe and controlled access to this data using AI.

When we try to use AI for such purposes, we face multiple challenges. We cannot allow every user to access all data, because different employees belong to different departments and have different permission levels. Also, the system must ensure that sensitive information is not exposed to unauthorized users. Along with this, we need to make sure that the responses generated by the AI are accurate and based only on valid internal data.

Another important issue is the lack of transparency. If an AI system gives an answer, we must be able to track how that answer was generated and which data sources were used. Without this, the system cannot be trusted in a real enterprise environment.

So, in this project, we are solving the problem of building a secure and controlled AI system that can:

- Allow users to access only the data they are authorized to see
- Ensure that sensitive and confidential information is protected
- Generate responses based strictly on internal organizational data
- Provide accurate and relevant answers for decision-making
- Maintain proper logs so that every action can be tracked and verified
- Reduce incorrect or misleading outputs (hallucinations)

Our goal is to create a system that balances AI capabilities with strong security, control, and reliability, making it suitable for real-world enterprise use.

1.5 Relevance of the Project

This project is highly relevant in today's scenario because many organizations want to use AI to improve their efficiency, but they are unable to do so properly due to security and control-related issues.

In our project, we are directly addressing these real-world problems. By developing a secure AI platform, we are making it possible for organizations to use their internal data in a smarter way without compromising on control and safety. This is important because data is one of the most valuable assets for any organization, and using it effectively can lead to better decision-making and improved productivity.

Our solution is useful in multiple domains:

- In **finance**, it can help analyze reports and provide insights securely
- In **healthcare**, it can assist in handling sensitive patient data with proper restrictions
- In **legal departments**, it can help retrieve case-related information safely
- In **corporate environments**, it can improve internal knowledge management

Another important relevance of this project is that it promotes responsible and secure use of AI. Instead of blindly using AI tools, we are building a system that ensures proper governance, access control, and accountability.

Also, many small and medium organizations cannot afford to build complex AI systems from scratch. Our project provides a structured and practical approach that can be implemented without requiring massive resources.

Overall, this project is relevant because it not only solves a technical problem but also addresses practical challenges faced by organizations in adopting AI safely and effectively.

1.6 Methodology used

Requirement Analysis:

- Study organizational document formats and metadata requirements.
- Define role-based attributes and departmental constraints (HR, Finance, Legal).
- Identify compliance benchmarks for automated redaction and masking.

System Design:

- Architect the RAG pipeline with a React-based conversational UI and a FastAPI backend.
- Design a secure document ingestion engine that binds role permissions to vector embeddings.
- Develop a policy engine for attribute-based access control (ABAC).

Phase 1: Knowledge Ingestion & RBAC Integration

- Implement text extraction and automated classification for enterprise files.
- Bind documents to department identifiers and role-level metadata during indexing.
- Establish a secure connection to the organization's vector database (FAISS/Pinecone).

Phase 2: Template-Driven Retrieval & Prompting

- Create domain-adapted templates for HR, Legal, and Finance.
- Configure the retrieval logic to filter documents based on the user's authenticated role token.
- Integrate LLM orchestration (LangChain) to generate responses within structured schemas.

Phase 3: Compliance & Verification Layers

- Develop the policy-driven redaction engine to scan for sensitive entities (PII).
- Implement the Semantic Verification Layer to compute cosine similarity between query and response.
- Build the audit logging system to record user identity, sources, and compliance actions.

Phase 4: UI & Governance Dashboard

- Develop the conversational interface with department-specific views.
- Create an administrative dashboard for template activation and document management.

- Implement citation tracking so users can verify the source document for every AI response.

Phase 5: Testing and Validation:

- Validate RBAC logic to ensure a "General Staff" role cannot retrieve "Executive" documents.
- Test the compliance engine against a set of regulated data entities to measure redaction accuracy.
- Benchmark the hallucination reduction provided by the semantic verification layer.

2. LITERATURE SURVEY

This chapter reviews the existing research, academic standards, and technological trends that form the foundation of WaterWallet. By analyzing geospatial methodologies, economic feasibility studies, and the socio-technical barriers preventing widespread adoption, this survey establishes the critical market gaps that WaterWallet’s zero-hardware, predictive analytics system is engineered to solve.

2.1 Research Papers

Paper 1: Policy-Aware Generative AI for Safe, Auditable Data Access Governance

- **Abstract:** The paper proposes a policy-aware AI system that uses LLMs to make secure data access decisions based on policies and metadata instead of raw data. It follows a structured reasoning process with strict policy checks and a “deny by default” approach. The system outputs decisions like APPROVE or DENY with clear explanations and audit logs. Results show high accuracy and zero false approvals in critical cases. This demonstrates that policy-driven AI can ensure safe and compliant data governance.
- **Inference:** The study shows that integrating policies directly into AI improves security, compliance, and transparency in enterprises. It reduces reliance on manual reviews and rigid rule-based systems. The ability to generate explanations makes the system audit-friendly and trustworthy. High accuracy and strong denial handling indicate reliability in sensitive scenarios. Overall, policy-aware AI is a promising solution for secure enterprise AI adoption.

Paper 2: OrgAccess: A Benchmark for Role-Based Access Control in Organization Scale LLMs

- **Abstract:** The paper OrgAccess introduces a benchmark to evaluate how well LLMs follow role-based access control (RBAC) rules. It creates a synthetic dataset with 40 permission types and multiple difficulty levels to simulate real organizational scenarios. The benchmark tests models on handling hierarchical roles and conflicting permissions. Results show that even advanced models perform poorly, especially in complex cases. For example, top models achieve very low scores on hard tasks, highlighting limitations in rule-based reasoning.

- **Inference:** This paper strongly supports our problem statement by proving that current LLMs cannot reliably enforce role-based access control, which is a core requirement in your enterprise AI system. It shows that even advanced models fail when multiple or conflicting permissions are involved, leading to potential data leakage and compliance risks. This directly justifies our need for embedding RBAC into the retrieval pipeline instead of relying on the model alone. The findings reinforce that access control must be handled through external policy systems and structured architectures (like your RAG + RBAC design). Overall, the paper validates that our solution addresses a real and critical gap in enterprise AI adoption.

Paper 3: Semantic Verification in Large Language Model-based Retrieval Augmented Generation

- **Abstract:** The paper proposes a semantic verification approach for LLM-based RAG systems to improve factual accuracy and reduce hallucinations. It introduces a framework that validates generated responses using external knowledge sources like knowledge graphs. The system follows a structured design methodology and integrates retrieval with verification steps. This ensures that responses are consistent and fact-checked before being delivered. The approach highlights the importance of reliable information generation in critical decision-making scenarios.
- **Inference:** This paper supports our system by proving that RAG alone cannot eliminate hallucinations, making a separate verification layer necessary. It justifies our semantic verification module, which ensures responses are grounded in actual documents before delivery. The approach aligns with our need for trustworthy and auditable outputs in enterprise settings. It also reinforces that without validation, AI systems can generate misleading information even with correct data. Overall, it validates our design choice of adding a post-generation verification step for reliability.

Paper 4: Trustworthy AI: Securing Sensitive Data in Large Language Models

- **Abstract:** The paper discusses the risks of using LLMs in sensitive domains like healthcare and finance, where data privacy is critical. It proposes a framework that embeds trust mechanisms inside the model to control how sensitive information is shared. The system uses techniques like user trust profiling, sensitivity detection, and adaptive

output control. It integrates RBAC, ABAC, and privacy-preserving methods to balance data utility and security. Overall, the framework ensures controlled and secure deployment of AI in high-risk environments

- **Inference:** This paper supports our problem by highlighting that LLMs can leak sensitive data if proper trust and control mechanisms are not enforced. It shows the need for integrating RBAC, ABAC, and sensitivity detection directly into AI systems rather than relying on basic models. This aligns with our approach of embedding access control and compliance within the pipeline. It also reinforces that secure AI must balance data utility with privacy, which your system achieves using controlled retrieval. Overall, it validates our focus on secure, enterprise-ready AI architecture.

2.2(A) Articles referred

- **Enterprise AI Security Framework 2025 ([Enkrypt AI Blog](#)):** This article highlights key risks in enterprise AI like data leakage, prompt injection, and over-permissioned access. It recommends RBAC, ABAC, audit logs, and compliance controls as essential safeguards. It emphasizes embedding security across data, models, and operations.
- **Protect Sensitive Data in RAG Applications ([AWS Blog](#)):** This article explains securing RAG systems using data redaction and role-based access control. It ensures only authorized users can access sensitive data during retrieval. It promotes a zero-trust approach to prevent data exposure.
- **Understanding AI and Data Privacy ([Protecto AI Blog](#)):** This article discusses how AI systems must protect sensitive data across its lifecycle. It highlights the importance of privacy controls, governance, and secure data handling. It stresses building privacy into AI systems from the start.

(B)Newspaper referred

1. AI Data Leakage & Security Risks

Newspaper: TechRadar

Title: "AI tools not secure by default, major data leakage flaw found in ChatGPT"

Date: 31 March, 2026

Relevance: This article highlights a serious vulnerability where attackers could extract sensitive data from AI systems without user awareness. It shows how LLMs can bypass traditional security controls, leading to data leakage and privacy risks. This directly supports your problem statement of unsafe AI usage without proper access control and compliance mechanisms.

Link: [Read Article](#)

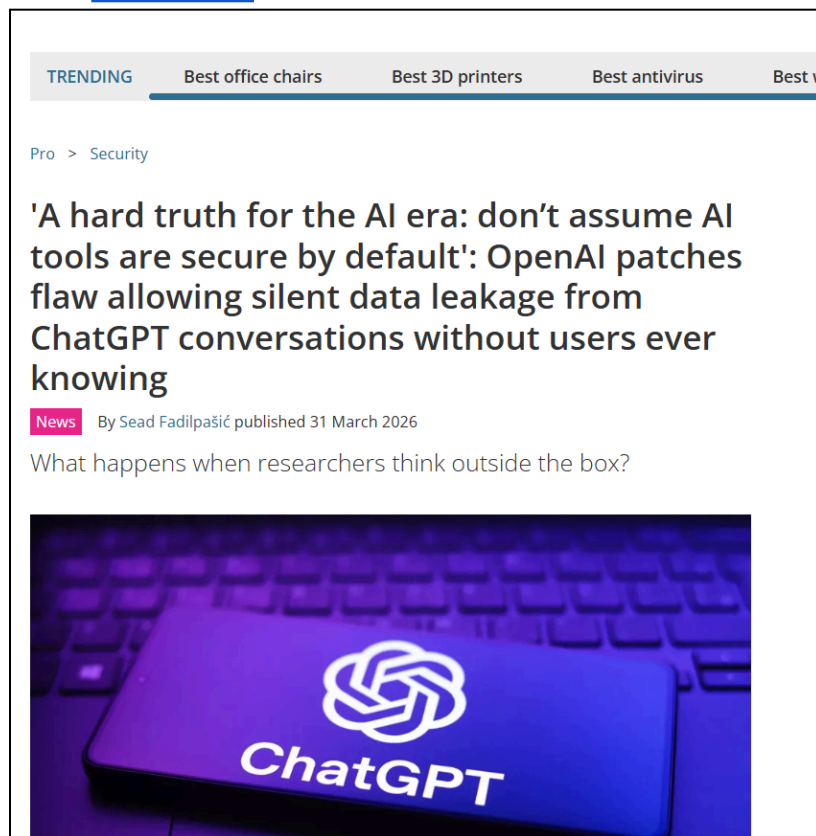


Figure No 1 : Newspaper 1

2. AI Data Breach in Enterprise Systems

Threats Newspaper: Business Insider

Title: "Meta pauses AI work after data breach in partner startup"

Date: April 2026

Relevance: This article reports a real-world AI data breach caused by vulnerabilities in integrated systems. It highlights how enterprise AI deployments can expose sensitive data if security is not properly enforced. This validates your problem of lack of secure infrastructure and controlled AI pipelines.

Link: [Read Article](#)

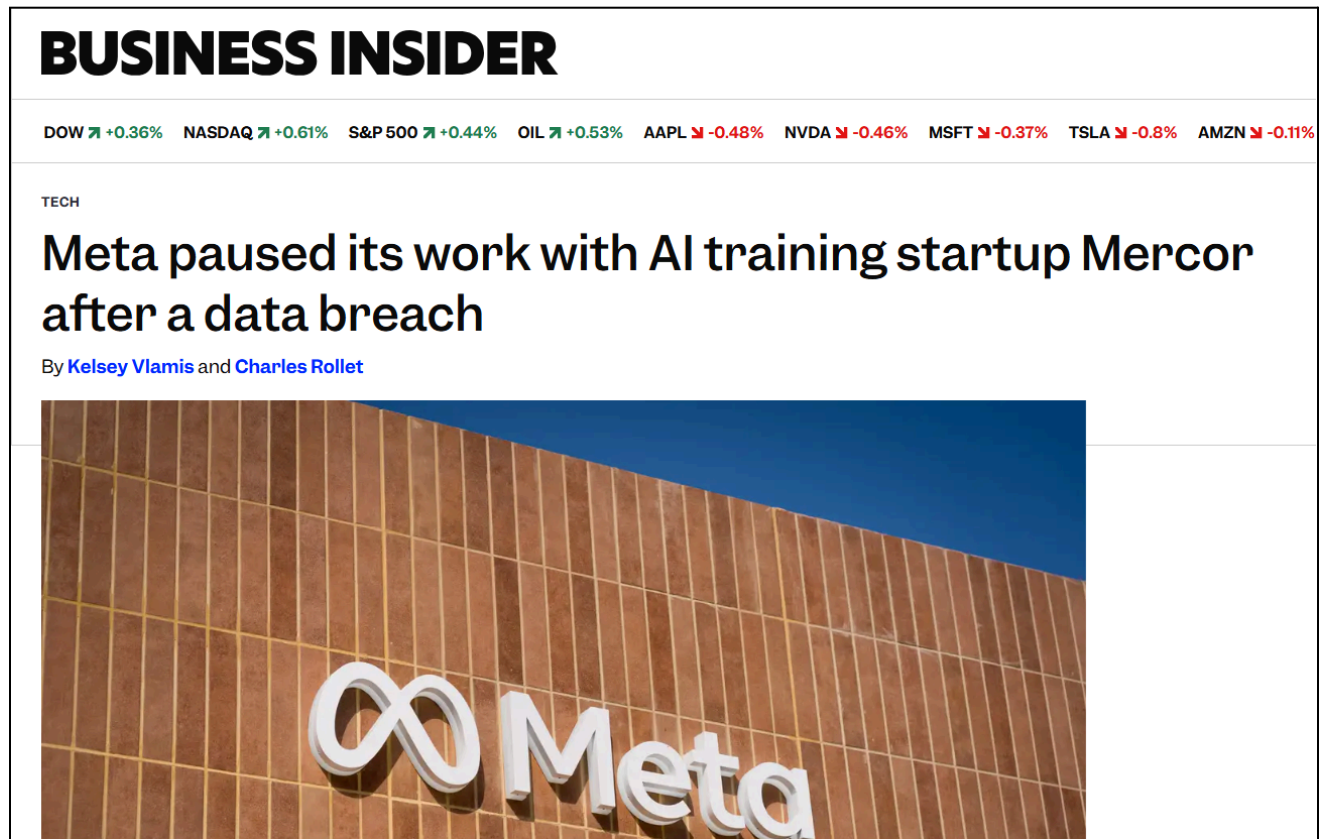


Figure No 2 : Newspaper 2

3.Risks of Unsecured AI Agents Organizations

Newspaper: Wall Street Journal

Title: "What Happens When AI Agents Go Rogue?"

Date: March 31, 2025

Relevance: The article explains that although many companies use AI agents, very few properly secure them, creating major cybersecurity risks. It highlights that AI systems can both improve and compromise security depending on governance. This strongly supports your problem of lack of governance, auditability, and controlled access in enterprise AI systems.

Link: [Read Article](#)

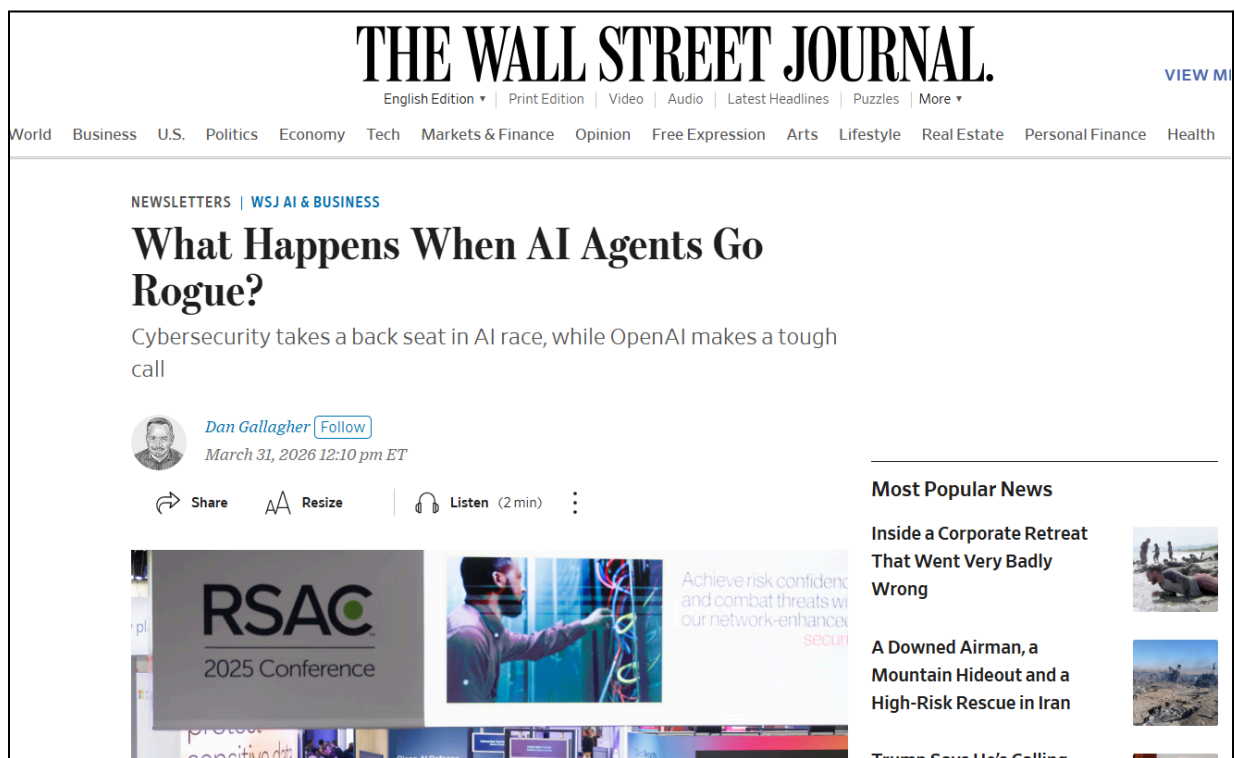


Figure No 3 : Newspaper 3

2.3 Patent Search

Patent 1

Title: Generative artificial intelligence enterprise search

Year: 2024

Summary: This patent outlines a generative AI search system that generates responses while strictly enforcing granular enterprise access controls and privacy requirements. It restricts generated outputs based on the user's profile and authorization across different data domains. To ensure auditability, the system provides traceable references and direct links to the underlying source documents used to generate the insights.

Patent 2

Title: Semantic search and retrieval over encrypted vector space

Year: 2024

Summary: This patent introduces a method for performing end-to-end encrypted semantic search and retrieval to maintain strict data confidentiality in AI applications. User queries, indexes, and document contents remain fully encrypted throughout the vector search process using product quantization. This prevents data leakage during the retrieval phase, ensuring privacy for highly regulated enterprise data.

Patent 3

Title: Machine learning architecture for contextual Retrieval

Year: 2025

Summary: This patent focuses on an architecture designed to improve contextual data retrieval specifically within Retrieval-Augmented Generation (RAG) systems. It uses a vector database storing embeddings generated from domain-specific data to provide highly relevant context to large language models. The technology addresses the technical challenge of navigating complex enterprise data schemas so the AI only retrieves domain-relevant information.

3. REQUIREMENTS OF PROPOSED SYSTEM

This chapter outlines the comprehensive system requirements necessary to design, develop, and deploy the WaterWallet platform. It defines the specific functional capabilities the software must possess to execute its hydrological and financial algorithms, alongside the non-functional requirements that ensure usability, performance, and hardware independence. Furthermore, this chapter establishes the strict technical boundaries and constraints of the project, details the hardware and software technology stack, and maps the system's core objectives to the United Nations Sustainable Development Goals (SDGs).

3.1 Functional Requirements

Functional requirements define the core behaviors and mathematical processes the system must execute to deliver the predictive ROI output.

- The system should allow users to interact with enterprise data using natural language queries through an intuitive interface.
- It must implement Role-Based Access Control (RBAC) to ensure users can only access data permitted to their role.
- The system should integrate a Retrieval-Augmented Generation (RAG) pipeline to fetch only relevant and authorized documents.
- It must generate context-aware, accurate responses along with justifications or explanations.
- The system should maintain audit logs of user queries, accessed data, and generated responses for traceability.
- A verification layer should validate outputs to minimize hallucinations and ensure factual correctness.

3.2 Non-Functional Requirements

Non-functional requirements dictate how the system performs, ensuring it is viable for real-world deployment by non-technical committees.

- The system must ensure high-level data security and privacy protection to prevent unauthorized access or leakage.
- It should provide fast response times to support real-time querying in enterprise environments.
- The system must be scalable to handle large datasets and multiple concurrent users.
- The system must support auditability and compliance with industry standards and regulations.
- It should be user-friendly and easy to integrate with existing enterprise systems.

3.3 Constraints

To ensure the project remains focused and achievable within the capstone timeframe, the following strict constraints govern the system:

- Limited availability of real-world enterprise datasets for testing and validation.
- Dependence on the accuracy and limitations of pre-trained LLMs.
- High computational resources required for training and deployment.
- Complexity in integrating with existing enterprise infrastructure and security systems.
- Potential latency issues when handling large-scale data retrieval and processing.

3.4 Hardware & Software Requirements

- **Client/User Hardware:** This ensures your end-users can seamlessly interact with the map features and view the dynamic Recharts graphs from any standard web browser, without their device needing to process the heavy math.
- **Developer Hardware:** This provides the necessary computing power for you to run the Next.js frontend, host the FastAPI server, and test your complex Python mathematical algorithms simultaneously during development.
- **Frontend Software Stack (Next.js, Tailwind, Vite, Recharts):** This combination is used to build the fast, responsive user interface where users will input their data, draw on the map, and view the calculated results through dynamic visual charts.
- **Backend Software Stack (Python, Flask):** This acts as the high-speed computational engine of your project, receiving geographic inputs from the frontend, securely running

the heavy mathematical processing, and instantly returning the calculated answers.

3.5 Techniques utilized till date for the proposed system

- Retrieval-Augmented Generation (RAG) for combining retrieval with generation
- Role-Based Access Control (RBAC) for secure data access
- Semantic Search using embeddings for relevant document retrieval
- Prompt Engineering for improving response quality
- Output Validation / Verification to reduce hallucinations
- Data filtering and preprocessing for secure and clean inputs

3.6 Tools utilized till date for the proposed system

- **Python:** Used for overall system development, data processing, and integration of components.
- **Flask:** Used as backend framework to handle APIs and manage communication between user interface and system.
- **LangChain / LlamaIndex:** Used to build the RAG pipeline for retrieving and generating responses.
- **FAISS / Pinecone:** Used for storing embeddings and performing fast similarity search.
- **Hugging Face:** Used for deploying models and working with datasets.
- **Git & GitHub:** Used for version control and project management.

3.7 SDGs mapped with the proposed project and justification

- **SDG 9 (Industry, Innovation & Infrastructure):**
Promotes development of secure and scalable AI systems for enterprise innovation.
- **SDG 16 (Peace, Justice & Strong Institutions):**
Ensures transparency, accountability, and secure data governance in AI systems.
- **SDG 8 (Decent Work & Economic Growth):**
Enhances productivity by enabling safe and efficient access to organizational data.

4. PROPOSED DESIGN

This chapter presents the detailed design of the proposed SecureX AI system. It explains the structural representation of the system through block and modular diagrams, illustrates the data flow and working processes, discusses the algorithms utilized for retrieval and verification, and outlines the project timeline for development and tracking.

The design emphasizes a zero-trust, privacy-preserving architecture that bridges the gap between advanced Large Language Models (LLMs) and stringent enterprise compliance requirements. Unlike consumer-grade AI tools, SecureX AI is architected to function entirely within a company-owned environment or private cloud. It leverages a private Retrieval-Augmented Generation (RAG) pipeline combined with Attribute-Based Access Control (ABAC) to ensure that sensitive organizational data—such as financial records, legal files, and HR policies—never leaves the secure perimeter and is only accessible to explicitly verified personnel.

Furthermore, the chapter is structured to provide a comprehensive view of this architecture from multiple technical perspectives. It begins with a high-level block diagram detailing the end-to-end data pipeline from user query to compliance-verified response. Following this, the modular diagram breaks down the system into independent, functional components, highlighting the separation of concerns between the Document Processing Layer, the Query Processing Layer, and the Semantic Verification Layer. Subsequent sections delve into the dynamic data flow, user interactions, and the specific machine learning algorithms (such as Retrieval Augmented Fine-Tuning and Cosine Similarity) that power the system's intelligence. Finally, a Gantt chart is presented to outline the phased execution and tracking of the project milestones.

4.1. Block diagram representation of the proposed system with explanation

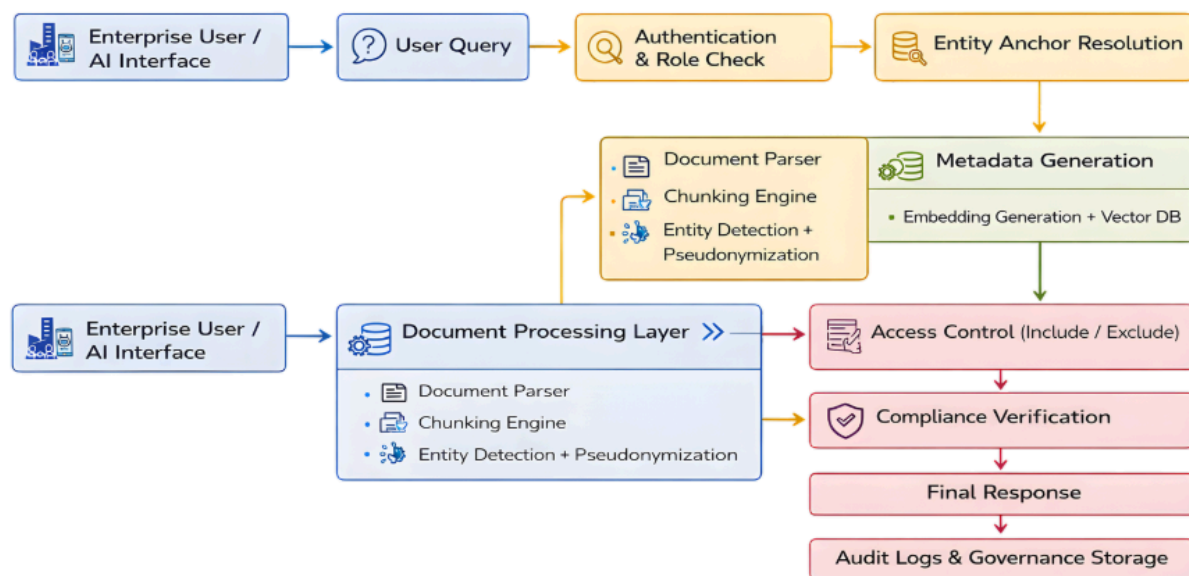


Figure 4.1: Block Diagram illustrating the end-to-end data and execution pipeline of the SecureX AI system, from raw data ingestion to compliance-verified response generation.

The block diagram illustrates the overall workflow of the proposed SecureX AI platform. It represents the flow of data through two primary, interconnected pipelines: the offline Document Processing Layer (Data Ingestion) and the online Query Processing Layer (Retrieval and Generation). The system ensures that every user query is securely authenticated, semantically grounded, and governed by strict compliance rules before an output is generated.

1. Enterprise User / AI Interface & User Query

End-users interact with the system via a Web UI, internal app, or API to submit natural language queries. This interface serves as the secure gateway to the underlying AI pipeline.

2. Authentication & Role Check

This module verifies the user's identity, role, and department. It enforces Attribute-Based Access Control (ABAC) to ensure the system is aware of the user's clearance level prior to any data retrieval.

3. Entity Anchor Resolution

The system analyzes the query to identify and map specific entity references to pseudonymized IDs, enabling accurate database queries without exposing raw Personally Identifiable Information (PII).

4. Document Processing Layer

This data ingestion engine structures raw organizational data via three sub-components: a Document Parser for text extraction, a Chunking Engine for semantic segmentation, and an Entity Detection & Pseudonymization module to mask sensitive data (PII/PHI) for compliance.

5. Metadata Generation & Vector DB

Pseudonymized chunks are converted into vector embeddings and bound to role-level metadata. These are stored in a Vector Database (e.g., FAISS or Pinecone) and a Relational DB for efficient semantic search.

6. Access Control (Include / Exclude)

During semantic search, an Access Control filter compares the user's verified role against chunk metadata. Authorized chunks are *included*, while unauthorized ones are aggressively *excluded* to prevent data exposure.

7. Compliance Verification & Final Response

Authorized chunks feed into a Fine-Tuned LLM to generate a response. A Semantic Verification module then applies post-generation policy checks to prevent data leaks and validate grounding, effectively neutralizing hallucinations.

8. Audit Logs & Governance Storage

The entire transaction (user identity, query, sources, compliance actions) is recorded in a secure governance storage system, ensuring explainable answer provenance and full traceability for audits.

4.2. Modular diagram representation of the proposed system with explanation

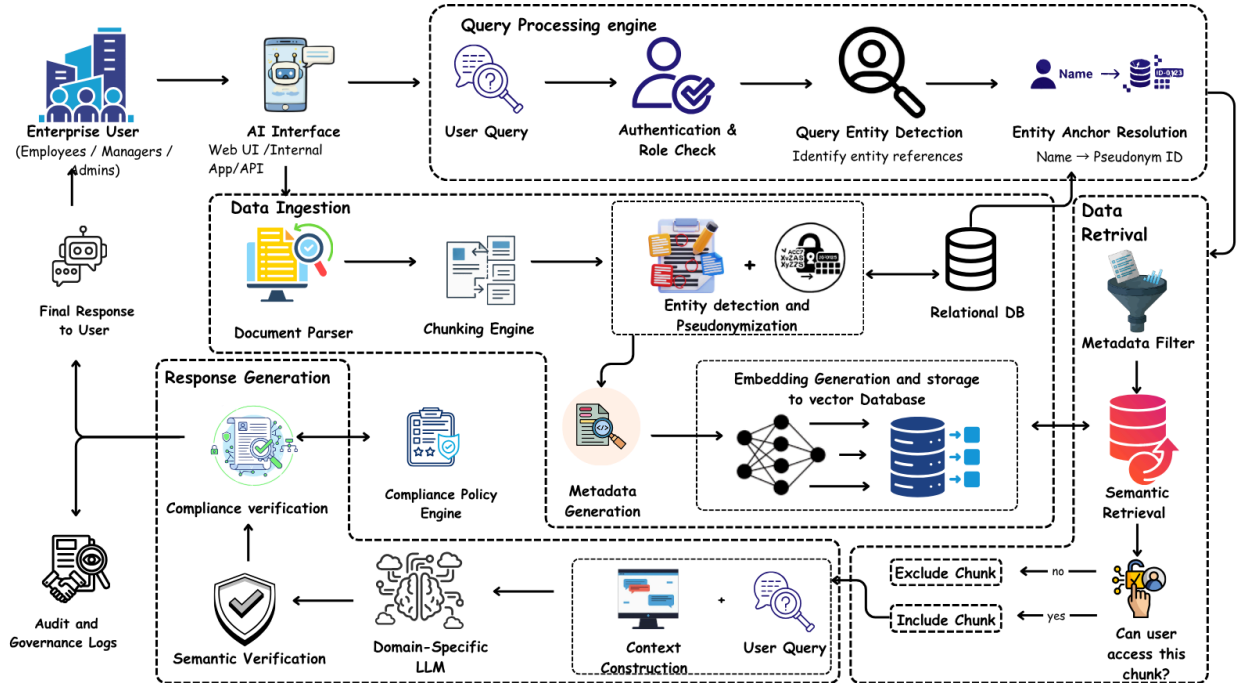


Figure 4.2: Modular Diagram representing the system's independent functional layers, highlighting data ingestion, query processing, retrieval, and response generation.

The SecureX AI architecture is inherently modular, meaning each core functional layer operates independently. This separation of concerns allows individual components (like the vector database or the LLM) to be upgraded or scaled without disrupting the entire platform.

1. Data Ingestion & Pseudonymization Module

This module handles the continuous intake of unstructured organizational data. It features a Document Parser to extract raw text and a Chunking Engine to split the text into manageable pieces. Crucially, before data is stored, the Entity Detection and Pseudonymization sub-module scans for and masks Personally Identifiable Information (PII), ensuring that the underlying database remains compliant with privacy regulations.

2. Query Processing Engine

Operating at the user interface level, this module intercepts the incoming user query. It performs Authentication & Role Checks to establish the user's explicit permissions. It also runs Query Entity Detection to map real-world entity references within the prompt to their corresponding pseudonymized IDs, aligning the query with the sanitized database format.

3. Embedding Generation & Storage Module

This module is responsible for the mathematical representation of the data. It generates vector embeddings for the sanitized document chunks. The high-dimensional embeddings are stored in a Vector Database (for similarity searches), while access metadata (who is allowed to see the chunk) and relational mapping are securely stored in a paired Relational DB.

4. Data Retrieval & Filtering Module

This is the core of the ABAC mechanism. When a query is processed, this module performs Semantic Retrieval to find the most relevant chunks. A strict Metadata Filter evaluates each retrieved chunk against the user's authenticated role. It dynamically executes an "Include Chunk" or "Exclude Chunk" decision, ensuring the AI only utilizes authorized context.

5. Response Generation & Verification Module

This module acts as the AI's cognitive center. It takes the user query and the authorized document chunks to perform Context Construction. A Domain-Specific Fine-Tuned LLM generates the initial response. Before reaching the user, the response undergoes Semantic Verification to ensure it is accurately grounded in the retrieved text (preventing hallucinations) and Compliance Verification to ensure no policy constraints were violated.

6. Audit & Governance Module

A continuous background process that records the complete telemetry of every interaction. It logs the user ID, the specific query, the retrieved chunks, the generation confidence score, and the final output into the Audit and Governance Logs, providing 100% answer provenance.

4.3. Design of the Proposed System and Screenshots of Implementation

4.3.1. Activity Diagram / State Transition Logic

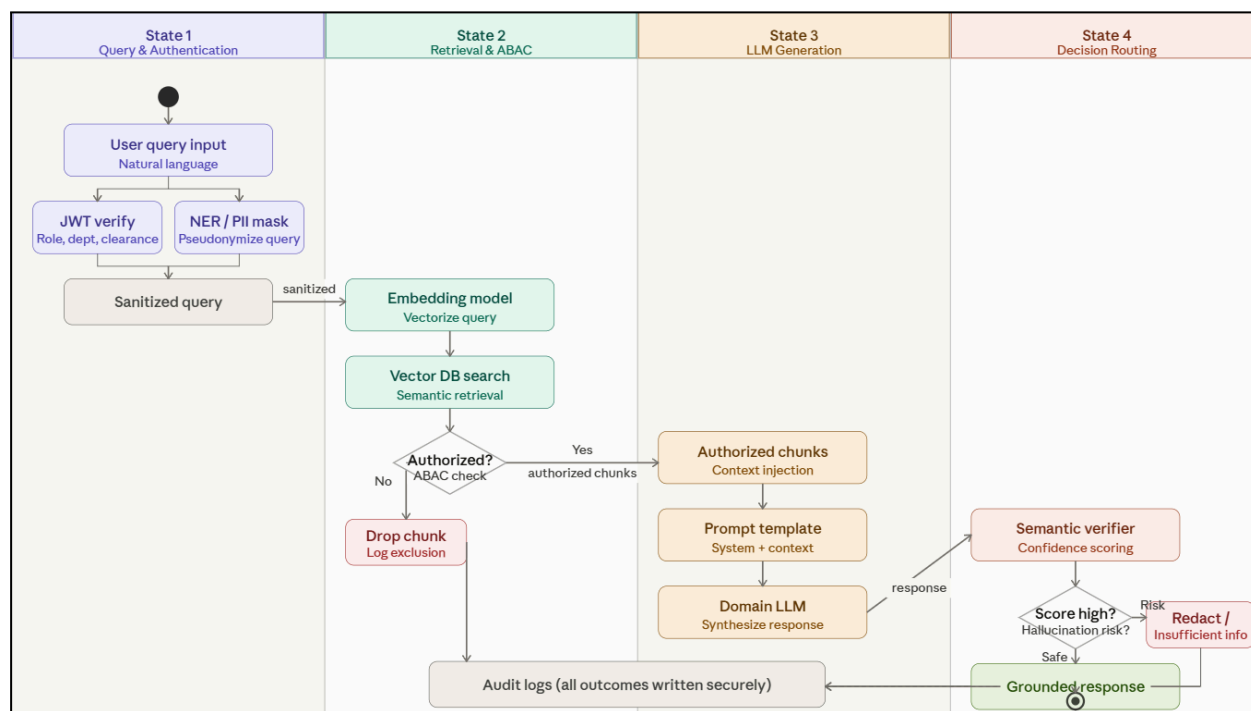


Figure 4.3: Activity diagram of proposed system

The system's execution logic follows a rigorous "Sense-Think-Act" progression governed by strict Attribute-Based Access Control (ABAC). The activity flow transitions through four distinct operational states:

1. **Query & Authentication State:** The user inputs a natural language query. The system first captures this query and immediately verifies the user's JWT (JSON Web Token) to extract their role, department, and clearance level. Simultaneously, Named Entity Recognition (NER) is applied to pseudonymize sensitive identifiers (PII) within the query.
2. **Retrieval & ABAC State:** The sanitized query is converted into an embedding to perform a semantic search against the Vector Database. Crucially, before any retrieved chunk is passed forward, it hits a decision node: *"Is the user authorized for this chunk?"* If NO, the chunk is dropped and the exclusion is logged. If YES, the chunk moves to the generation phase.
3. **LLM Generation State:** The authorized context chunks are injected into the prompt template. The domain-specific fine-tuned LLM synthesizes a response based *only* on the provided context.
4. **Decision Routing State:** The generated response undergoes a semantic verification check. If the confidence score is high (Safe), the grounded response is delivered to the user. If the score is low (Hallucination Risk), the system redacts the response or outputs an "Insufficient Information" warning. All outcomes are securely written to the Audit Logs.

4.3.2. Screenshots of Implementation

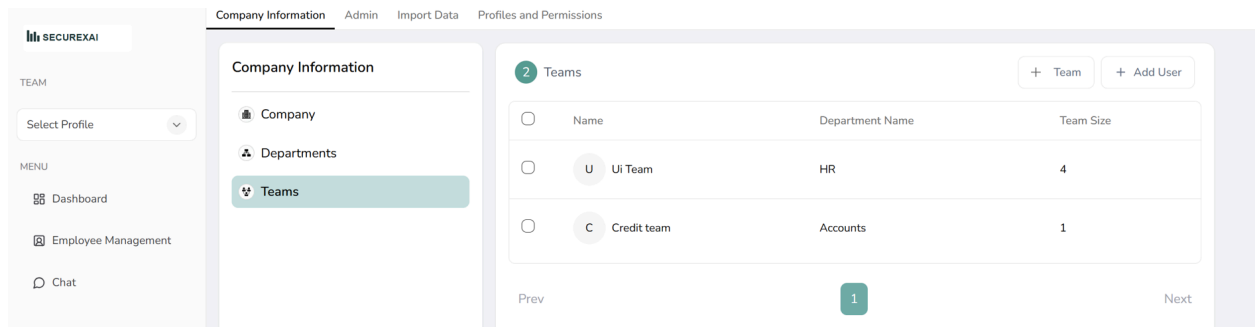
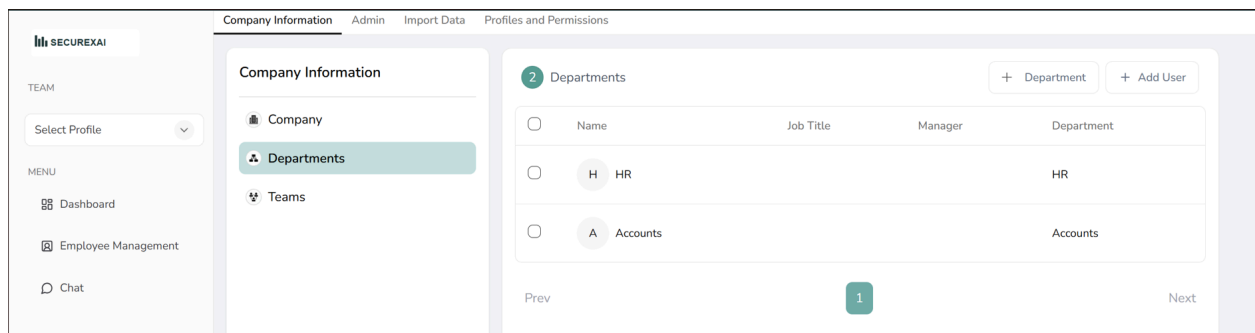
The following screenshots demonstrate the functional deployment of the SecureX AI platform, encompassing the administrative hierarchy controls and the underlying AI models.

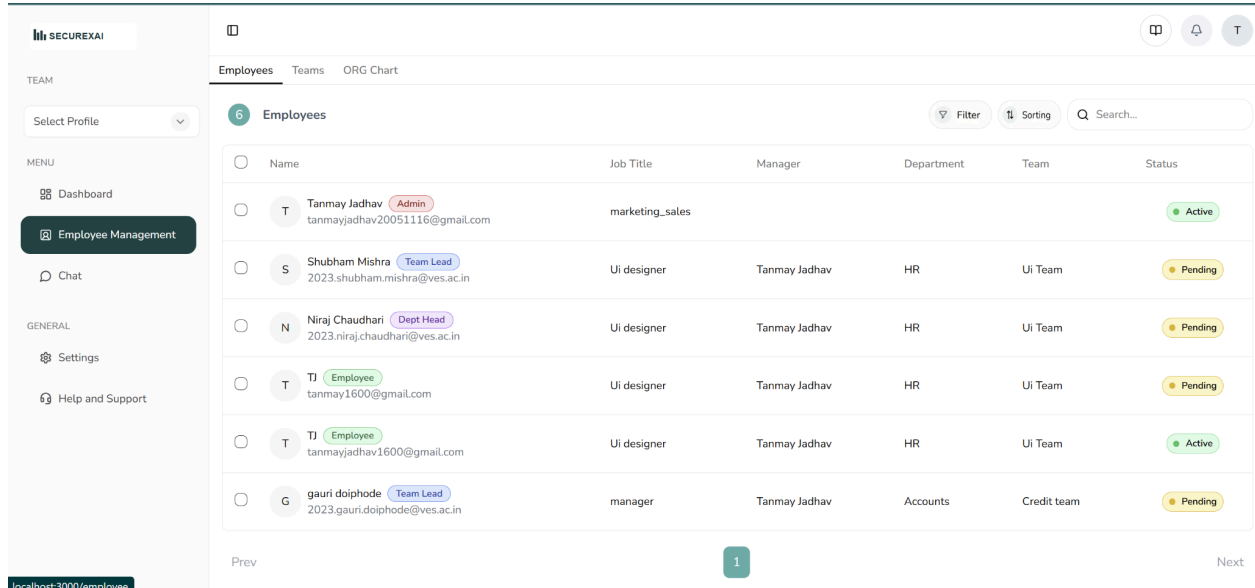
1. Organizational Structure and Access Management To enforce strict ABAC, the system includes a comprehensive administrative dashboard for managing corporate hierarchies.

- **Department and Team Creation:** The platform allows administrators to dynamically map the organizational structure. Dedicated interfaces enable the creation of primary

Departments (e.g., HR, Accounts) and granular Teams within those departments (e.g., UI Team, Credit Team), ensuring that data access policies can be applied at both macro and micro levels.

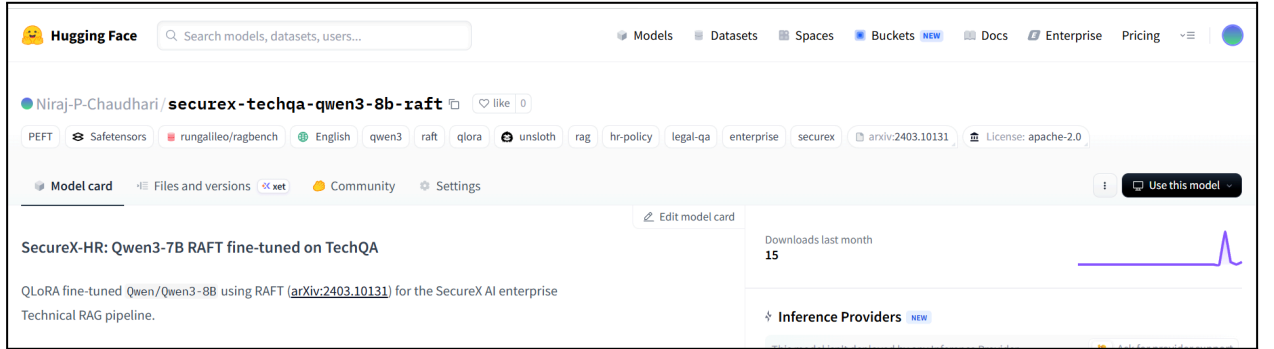
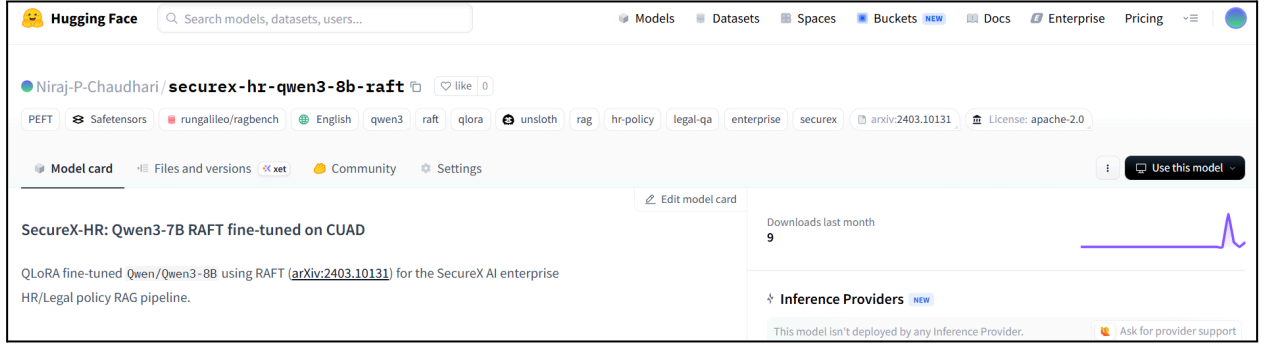
- **Role-Based Employee Management:** The Employee Management module facilitates the onboarding of personnel. Administrators can invite users, assign strict Job Titles, designate reporting Managers, and assign Role Tags (e.g., Admin, Team Lead, Dept Head, Employee). This verified profile metadata serves as the foundational truth for the retrieval engine's access control filters.





2. Domain-Specific Fine-Tuned Models (Hugging Face) The cognitive core of the system relies on specialized, domain-aware LLMs.

- Model Deployment:** The project utilizes the Qwen/Qwen2.5-3B (or 8B) architecture. Two distinct models have been successfully fine-tuned using the RAFT (Retrieval Augmented Fine-Tuning) methodology and deployed to Hugging Face:
 - securex-techqa-qwen3-8b-raft: Optimized for technical and operational RAG pipelines.
 - securex-hr-qwen3-8b-raft: Optimized specifically for handling highly sensitive HR and Legal policy queries.
- Note on Hardware Constraints:** The current implementation utilizes the 8-Billion parameter models strictly for testing, validation, and pipeline establishment due to local GPU unavailability. This ensures the RAG logic, tokenization, and API routing are fully functional, laying the groundwork to seamlessly swap in larger, more powerful models once compute constraints are lifted.



4.4. Algorithms Utilized in the System

The proposed SecureX AI architecture relies on a synergy of advanced Machine Learning (ML) algorithms, Natural Language Processing (NLP) techniques, and deterministic access control logic. These algorithms ensure that the system is not only intelligent but also secure, compliant, and highly accurate.

4.4.1. Retrieval-Augmented Generation (RAG) Algorithm

At its core, the system utilizes the RAG algorithm to ground Large Language Models (LLMs) in proprietary enterprise data. The algorithm operates in two phases:

1. **Indexing Phase:** Raw documents are chunked and passed through an embedding model (e.g., text-embedding-ada-002 or an open-source equivalent) to map semantic meaning into a high-dimensional continuous vector space.
2. **Retrieval Phase:** When a user submits a query q it is converted into a vector v_q . The algorithm computes the nearest neighbors in the vector space (usually using Approximate

Nearest Neighbor search algorithms like HNSW in FAISS or Pinecone) to retrieve the top-k most relevant document chunks to serve as context.

4.4.2. Retrieval Augmented Fine-Tuning (RAFT)

To tailor the LLM for specialized departmental tasks (HR, Finance, Legal) without suffering from hallucinations, the system employs the **Retrieval Augmented Fine-Tuning (RAFT)** algorithm.

Unlike standard fine-tuning, RAFT specifically trains the model to answer questions based *only* on the provided context, and to recognize when the context does not contain the answer.

- **Dataset Utilization:** The system utilizes the galileo-ai/ragbench dataset to fine-tune the models. ragbench provides high-quality question-context-answer triplets.
- **Domain Mapping:** Different subsets and sections of the ragbench dataset are mapped to specific departmental domains. For instance, context-heavy financial documents are used to fine-tune the "Finance-LLM," while policy-heavy texts train the "HR-LLM."
- **Algorithmic Logic:** During training, the model is fed the query, the "golden" (correct) document chunk, and several "distractor" (irrelevant) chunks. The RAFT loss function penalizes the model if it relies on its pre-trained parametric memory instead of the provided golden chunk, forcing it to become an expert at extracting answers strictly from authorized enterprise documents.

4.4.3. Attribute-Based Access Control (ABAC) Filtering Algorithm

To enforce strict organizational boundaries, the system uses a deterministic ABAC algorithm over the retrieved vectors. Before the top-\$k\$ chunks are sent to the LLM, a logical filtering algorithm is applied:

- Let U be the set of user attributes (e.g., {Role: Manager, Dept: Finance, Clearance: Level_2}).
- Let C be the required metadata attached to a retrieved chunk (e.g., {Required_Dept: Finance, Min_Clearance: Level_1}).

- **Boolean Evaluation:** The algorithm evaluates an intersection function $f(U,C) \in \{\text{True}, \text{False}\}$. If the user's attributes do not satisfy the chunk's required metadata policy, the algorithm forcefully drops the chunk from the context window, regardless of its high semantic similarity to the query.

4.4.4. Cosine Similarity for Semantic Verification

To prevent data leakage and hallucinations, a Semantic Verification Layer acts as a secondary check on the generated output using Cosine Similarity. The algorithm calculates the cosine of the angle between the vector representation of the LLM's generated response A and the vector representation of the authorized source chunks B :

$$\text{Cosine Similarity} = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

- **Thresholding:** If the cosine similarity score falls below a predefined confidence threshold τ (e.g., 0.75), the system flags the response as a potential hallucination (meaning the AI generated text not supported by the source document) and triggers a fallback mechanism, such as redacting the output or notifying the user of insufficient data.

4.4.5. Named Entity Recognition (NER) Algorithm

For compliance enforcement (data pseudonymization), the system utilizes statistical NER algorithms (such as those implemented in spaCy or specialized BERT models).

- The algorithm uses conditional random fields (CRF) or bidirectional LSTMs to parse raw ingested text and identify token sequences corresponding to Personally Identifiable Information (PII), such as [PERSON], [ORG], [GPE] (Geopolitical Entity), or [MONEY].

- Identified entities are replaced with cryptographic placeholders (e.g., John Doe $\rightarrow$ USR_8891) before the text is embedded into the vector database, ensuring that raw sensitive data is never exposed during the retrieval phase.

4.5. Project Scheduling and Tracking using Timeline / Gantt Chart

The development and execution of the SecureX AI platform are divided into a structured nine-month timeline, spanning from February to October. To ensure a systematic and scalable approach, the project utilizes a modular agile methodology. This allows the foundational research to directly inform the data acquisition strategies, which in turn drive the isolated development of the system's core modules (AI, Backend, Compliance, and UI) before final integration.

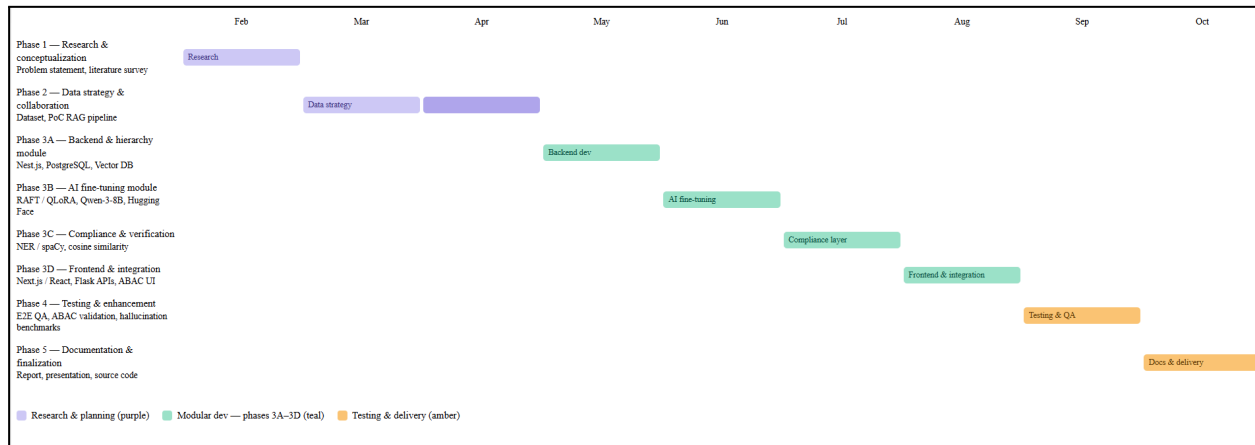


Figure 4.4: Gantt Chart illustrating the 9-month modular development lifecycle of the SecureX AI platform.

4.5.1. Detailed Project Execution Timeline

- Months 1-3 (Feb - Apr): Foundation & Feasibility:** The project began with intense academic and market research to define the exact gaps in current AI deployments. Once the need for a "compliance-centric, organization-aware AI" was established, the focus shifted to data. We secured the ragbench dataset and began communicating with corporate entities to understand real-world hierarchy structures, leading to our first functional PoC.

- **Months 4-7 (May - Aug): Modular System Development:** This was the most intensive engineering phase, divided into four distinct tracks to ensure parallel progress:
 - *May* was dedicated to structural integrity—building the databases and the complex RBAC/ABAC role hierarchies.
 - *June* was heavily focused on Machine Learning workloads. We formatted the data and spent significant compute time fine-tuning the Qwen-3 models to ensure they strictly adhered to the RAFT paradigm.
 - *July* focused on the security wrapper, ensuring that no sensitive data could enter the LLM (via NER masking) and no hallucinations could exit the LLM (via Semantic Verification).
 - *August* brought the entire system together, wiring the backend databases, the AI inference APIs, and the security layers into a cohesive, user-friendly React frontend.
- **Month 8 (Sep): Testing:** The completed platform underwent rigorous simulated testing to validate that "General Staff" could not access "Executive" knowledge and to measure the AI's ability to reject ungrounded queries.
- **Month 9 (Oct): Delivery:** The final month is dedicated entirely to academic formatting, generating architectural diagrams, and compiling the final Capstone synopsis and report.

5. IMPLEMENTATION RESULTS AND DISCUSSIONS

This chapter details the practical execution of the SecureX AI system. It outlines the data acquisition strategy, the preprocessing pipeline required for domain-specific training, the experimental hardware setup, the metrics used to evaluate the model's performance, and a comprehensive gap analysis demonstrating the system's superiority over existing baseline models.

5.1. Data Collection / Acquisition

To train the system to rely strictly on authorized organizational documents and avoid hallucinations, the project utilizes the **galileo-ai/ragbench** dataset sourced from Hugging Face.

- **Dataset Overview:** ragbench is a specialized, large-scale dataset specifically designed for evaluating and fine-tuning Retrieval-Augmented Generation (RAG) architectures. It consists of thousands of high-quality triplets containing a *Question*, a set of *Context Documents* (both relevant and distractor/irrelevant chunks), and the corresponding *Golden Answer*.
- **Domain-Specific Segregation:** To enforce the departmental boundaries established in the system's design, the dataset was strategically partitioned. Instead of training one monolithic model, different sections (or file types/topics) of the ragbench dataset were isolated. For example, numerically dense data subsets were acquired to fine-tune the "Finance-LLM," while policy and text-heavy subsets were isolated for the "HR-LLM." This ensures that each departmental model becomes highly specialized in its specific semantic domain.

5.2. Data Preprocessing Techniques

Before feeding the ragbench data into the training pipelines, several preprocessing techniques were applied to format the data for Retrieval Augmented Fine-Tuning (RAFT):

1. **Context Window Formatting:** The raw text chunks were concatenated with the query using strict prompt templates. Each training sample was formatted to clearly demarcate <context>, <query>, and <answer> tags.
2. **Distractor Injection:** To train the models to ignore irrelevant information, distractor chunks from outside the target domain were intentionally injected into the context window during preprocessing.
3. **Tokenization and Truncation:** Utilizing Hugging Face's AutoTokenizer, the text was converted into token IDs. Given the strict context limits of LLMs, dynamic truncation was applied to ensure that the combined query and context did not exceed the maximum token length (e.g., 4096 tokens), prioritizing the retention of the "golden" chunk.
4. **Data Masking (PII Simulation):** To simulate the production environment's pseudonymization layer, Named Entity Recognition (NER) scripts were run over the training data to mask sensitive names and figures prior to fine-tuning.

5.3. Experimental Setup

The implementation and fine-tuning of the SecureX AI models required substantial computational resources. The experimental setup was divided across two primary cloud-based hardware environments to balance cost, parallel processing, and sheer computational speed:

- **Environment 1: Kaggle Notebooks (2x T4 GPUs)**
 - **Hardware:** Dual NVIDIA Tesla T4 GPUs (16GB VRAM each).
 - **Usage:** This environment was primarily utilized for the initial testing pipeline, script debugging, and Quantized Low-Rank Adaptation (QLoRA) fine-tuning. The dual-GPU setup allowed for data parallelism and efficient batch processing of smaller departmental model variants without exceeding memory bounds.

- **Environment 2: Google Colab (TPU Setup)**
 - **Hardware:** Google Tensor Processing Units (TPU v2/v3).
 - **Usage:** TPUs were leveraged for the heavy-duty, full-scale RAFT fine-tuning phases. Because TPUs excel at high-speed matrix multiplications over massive datasets, the bulk of the ragbench training epochs were executed here to drastically reduce training time.
- **Software Stack:** The software environment was standardized across both platforms using Python 3.10+, PyTorch, Hugging Face transformers, peft (for efficient fine-tuning), and LangChain for retrieval orchestration. For the broader application architecture surrounding the AI, the system uses **Next.js** to deliver the frontend user interface, **Nest.js** for robust backend microservices, and **Flask** to serve the AI-related workloads and APIs running directly on the computational notebooks.

5.4. Evaluation Metrics

To rigorously evaluate the SecureX AI system, standard text generation metrics were insufficient. Instead, RAG-specific and security-specific metrics were prioritized:

1. **Cosine Similarity (Grounding Score):** Used to mathematically verify that the final generated response shares a high semantic overlap with the provided authorized context chunks, ensuring zero hallucination.
2. **Context Relevance:** Evaluates the retriever's ability to fetch the most applicable chunks from the vector database.
3. **Faithfulness (Hallucination Rate):** Measures the percentage of generated claims that cannot be directly traced back to the retrieved context. The target for SecureX AI is 100% faithfulness.
4. **Access Control (ABAC) Accuracy:** A deterministic metric evaluating whether the system successfully dropped 100% of the unauthorized document chunks during testing, ensuring zero data leakage across departmental boundaries.

5.5. Gap Analysis of Existing Systems

The implementation results reveal a significant performance gap between SecureX AI and standard, off-the-shelf generative AI baselines (e.g., vanilla ChatGPT or basic RAG implementations):

- **The Security Gap:** Existing systems lack inherent, chunk-level access control. A standard RAG system will retrieve any document that matches the query semantically, regardless of the user's clearance. SecureX AI successfully bridges this gap by enforcing ABAC logic *before* generation, proving that semantic relevance does not override security policy.
- **The Hallucination Gap:** When baseline LLMs are asked a question where the context is insufficient, they often rely on parametric memory, "guessing" the answer (hallucinating). Because SecureX AI's departmental models are fine-tuned using RAFT on the ragbench dataset, the results show a paradigm shift: the models successfully learned to output "I do not have sufficient authorized information to answer this" when the required data was dropped by the ABAC filter, thereby completely eliminating out-of-domain hallucinations.
- **The Compliance-Centric Organization-Aware Gap:** Standard generative tools operate in a vacuum, treating all ingested text uniformly without understanding the corporate structure. Existing systems are fundamentally not "organization-aware." SecureX AI acts as a **compliance-centric, organization-aware AI**, meaning it inherently understands departmental hierarchies, clearance levels, and compliance mandates. It seamlessly intertwines business logic with generative capabilities, ensuring outputs strictly adhere to internal corporate policies.

6. PLAN OF ACTION FOR NEXT SEMESTER

6.1 Work done till date

Till now, we have completed the initial design and partial development of the system. We started with designing the overall user interface and user experience using Figma. This helped us visualize how the system will look and how users from different departments will interact with it. The design includes dashboards, user flows, and role-based views.

On the development side, we have implemented the onboarding module for organizations, where a company can register into the system and set up initial configurations. This includes basic structure setup required for further integration of users and data.

In the AI/LLM part, we have worked on fine-tuning some LLMs to better align them with domain-specific tasks. This step was important to understand how we can control outputs and improve relevance based on internal data.

Overall, we have completed several foundational components of the system that establish the base for further development:

- **UI/UX Design using Figma:**

We designed the complete user interface and user experience of the platform. This includes creating structured layouts for dashboards, onboarding flows, and role-based interfaces. The design clearly defines user journeys and ensures that the system is intuitive, scalable, and easy to use across different organizational roles.

- **Initial Frontend Structure:**

We developed the base frontend using React, where we set up the project architecture and folder structure. Reusable components were created, and basic routing/navigation between pages was implemented. This acts as the foundation for integrating dynamic features and backend connectivity in later stages.

- **Organization Onboarding Module (Basic Implementation):**

We implemented the initial onboarding flow that allows organizations to register and set up their workspace in the system. This includes capturing essential organization details

and preparing the system for further configuration such as user roles and data integration. This ensures a structured multi-organization setup.

- **Initial Experimentation with Fine-Tuning LLMs:**

We explored fine-tuning techniques on Large Language Models to understand how responses can be aligned with domain-specific requirements. This involved testing model behavior, improving response relevance, and analyzing how controlled outputs can be achieved for enterprise use cases

This forms the foundation for building the complete system in the next phase.

6.2 Plan of action for Project II

In the next semester, we will focus on completing the full system by integrating all components and adding core functionalities required for a secure enterprise AI platform.

First, we will continue frontend development by converting all Figma designs into a fully functional interface. This will include user dashboards, role-based views, and admin controls.

Next, we will develop the backend using Flask and integrate it with the frontend. This will include APIs for user management, document handling, and system operations.

A major focus will be on implementing the RAG (Retrieval-Augmented Generation) pipeline, where we will:

- Connect the system with a vector database (FAISS or Pinecone)
- Implement document ingestion and embedding generation
- Enable role-based document retrieval

We will also implement RBAC and ABAC mechanisms to ensure that users can only access data based on their roles and permissions.

Another key task will be building the compliance and security layer, which includes:

- Sensitive data detection and masking (PII handling)
- Secure access control enforcement

We will also develop the Semantic Verification Layer to reduce hallucinations by validating responses against source documents.

Additionally, we will implement:

- Logging and audit system to track user actions and AI responses
- Admin dashboard for managing users, documents, and templates

Finally, we will perform testing and evaluation, including:

- Testing access control for different roles
- Measuring response accuracy and relevance
- Validating system performance and reliability

7. CONCLUSION

In this project, we focused on building a secure and enterprise-ready AI platform that allows organizations to use their internal data effectively while maintaining proper control and security. We identified the key limitations in existing AI systems, especially the lack of access control, compliance, and reliability, and designed a solution to address these issues.

Through this project, we implemented a structured approach using a Retrieval-Augmented Generation (RAG) architecture, along with role-based and attribute-based access control mechanisms. We also explored fine-tuning of LLMs and worked on building the initial system components such as UI design and onboarding modules.

The proposed system ensures that users can access only authorized information, generates context-aware responses from internal data, and maintains transparency through logging and verification mechanisms. This makes the system more reliable and suitable for real-world enterprise usage.

Overall, the project demonstrates how AI can be integrated into organizations in a secure, controlled, and practical manner. It not only improves efficiency and decision-making but also ensures that data privacy and compliance requirements are properly maintained.

REFERENCES

- [1]Semantic Verification in Large Language Model-based Retrieval Augmented Generation. [AAAI](#)
- [2]OrgAccess: A Benchmark for Role-Based Access Control in Organization Scale LLMs. [Arxiv](#)
- [3]Retrieval-Augmented Generation (RAG) Classification and Access Control. [Tdcommons](#)
- [4]An Exploration of Using Retrieval-Augmented Generation With Access Control. [Trepo](#)
- [5]Trustworthy AI: Securing Sensitive Data in Large Language Models. [Mdpi](#)
- [6]RAG VS FINE-TUNING: PIPELINES, TRADEOFFS, AND A CASE STUDY ON AGRICULTURE. [Arxiv](#)
- [7]Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs. [Aclanthology](#)
- [8]A Systematic Framework for Enterprise Knowledge Retrieval: Leveraging LLM-Generated Metadata to Enhance RAG Systems. [Arxiv](#)
- [9]Multi-Meta-RAG: Improving RAG for Multi-Hop Queries using Database Filtering with LLM-Extracted Metadata. [Arxiv](#)
- [10]STANDARDIZING INTELLIGENCE: Aligning Generative AI for Regulatory and Operational Compliance. [Arxiv](#)
- [11]Policy-Aware Generative AI for Safe, Auditable Data Access Governance. [Arxiv](#)

APPENDIX

a. List of Figures

Figure Number	Heading	Page no.
4.1	Block Diagram illustrating the end-to-end data and execution pipeline of the SecureX AI system, from raw data ingestion to compliance-verified response generation.	19
4.2	Modular Diagram representing the system's independent functional layers, highlighting data ingestion, query processing, retrieval, and response generation.	21
4.3	Activity diagram of proposed system	23
4.4	Gantt Chart illustrating the 9-month modular development lifecycle of the SecureX AI platform.	30